

Infineon Linux Bluetooth® release notes

About this document

Scope and purpose

This document offers a summary of the Bluetooth® firmware release, highlighting known bug fixes and updated specifications specific to the AIROC™ combo chip (CYW55513/2/1, CYW55573/2/1, CYW43022, CYW43439, CYW43455 and CYW4373) in the latest Linux release. It also includes information on the proprietary Bluetooth® Stack (AIROC™ Bluetooth® Stack) and related code examples.

Intended audience

This document is intended for anyone who uses the AIROC™ Wi-Fi & Bluetooth® combo chips on a Linux host.

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Overview

1 Overview

1.1 High-level summary

This document details the new features and bug fixes done for CYW55513/2/1, CYW55573/2/1, CYW43022, CYW43439, CYW43455, and CYW4373 on the Bluetooth® Linux software release.

This software release includes the following:

- Compliance with the latest Bluetooth® specifications
- Bug fixes
- Bluetooth® stack updates
- Code example updates

1.2 Device firmware revision details

Table 1 Bluetooth® firmware revision details

Device	Bluetooth® firmware version	Bluetooth® version	QDID
CYW55513/2/1	001.002.032.0180.0000	6.0	Q338655
CYW55573/2/1	001.002.087.0356.0000	6.0	Q345126
CYW43022	003.002.024.0040.0000	5.4	207748
CYW43439	001.003.016.0072.0000	5.4	222970
CYW43455	003.001.025.0196.0000	5.3	183141
CYW4373	001.001.025.0130.0000	5.4	228194

Note: AIROC™ Bluetooth® Stack v3.9.2 is validated on the Linux Kernel v 6.6.31 with Raspberry Pi CM4.

Firmware changes

2 Firmware changes

This section describes the firmware updates, including bug fixes and enhancements, which may impact various AIROC™ devices.

2.1 CYW55513/2/1

CYW55513, CYW55512, and CYW55511 are designed to address the needs of the Internet of Things (IoT) devices that require minimal power consumption and compact size.

- CYW55513/CYW55512/CYW55511 support Bluetooth®/Bluetooth® Low Energy (LE)
- Bluetooth® 6.0 certified firmware
- Enhancement of the isochronous adaptation layer (ISOAL), a Bluetooth® 6.0 feature
- LE audio
 - LE audio with LC3 codec on the device (Offload mode)
 - LE audio with LC3 codec on the host (HCI mode)
- LE isochronous channels
- Bluetooth® 5.0/5.1/5.2/5.3/5.4/6.0 features:
 - LE long range (LR)
 - LE 2 Mbps, LE 1 Mbps
 - LE mesh
 - Advertising extensions

2.1.1 Bug fixes

- Improved LE Audio quality
- Bug fix for MIC failure in Dual Unicast LE Audio Stream
- Bug fix for MIC failure with LE Audio IOP device

Firmware changes

2.1.2 Known issues

Table 2 Known issues for CYW55513/2/1

Supported features	Known issues
Coding Scheme Selection for Advertising (CSSA)	None
Periodic advertising ADI support, LE Channel Classification, and LE Enhanced Connection Update	LE CIS connection establishment fails when the corresponding LE-ACL connection is subrated.
LE Core Isochronous channels, LE Power Control	Disconnection due to MIC failure occurs in certain scenarios with multiple LE-CIS links. Disconnection due to connection timeout is observed in certain scenarios with IUT as peripheral in single LE-CIS. Auracast synchronizer intermittently fails to establish sync to Auracast Broadcaster.
Periodic Advertising Sync transfer	None
Extended Advertising feature, 2 Mbps LE	None
LE Privacy, Data packet length extension (DPLE)	None

2.2 CYW55573/2/1

2.2.1 Enhancement

- Bluetooth® 6.0 certified firmware
- Enhancement of ISOAL, (Bluetooth® 6.0 feature)
- Added support for multiple connectable Extended Advertising sets

2.2.2 Bug Fixes

- Bug fixes and improvement in LE ISOC
- Bug fix for APCF filtering for 32-bit and 128-bit UUID service data

2.2.3 Known issues

- Bluetooth® ACL disconnection occurs when using EV3 on eSCO and 2/3 Mbps packet on Bluetooth® ACL, as IUT does not respond to 2/3 Mbps data packets on Bluetooth® ACL
- Auracast synchronizer intermittently fails to establish sync to Auracast Broadcaster
- Disconnection due to MIC failure occurs in certain scenarios with IUT as Central and Audio Sink in single unicast LE Audio Stream

2.3 CYW43022

2.3.1 Bug fixes

- Fixed packet counter issue that causes LE connection drop with reason MIC failure
- Fixed filter list not correctly programmed issue after controller exits Sleep mode

Firmware changes

2.3.2 Known issues

- No issues reported during internal sanity testing

2.4 CYW43439

2.4.1 Bug fixes

- Fixed bug affecting SCO connection stability

2.4.2 Known issues

- No issues reported during internal sanity testing

2.5 CYW43455

2.5.1 Bug fixes

- BT 5.3 updates

2.5.2 Known issues

- No issues reported during internal sanity testing

2.6 CYW4373

2.6.1 Bug fixes

- Improved eSCO connection reception

2.6.2 Known issues

- No issues reported during internal sanity testing

2.7 BTSTACK updates - v3.9.2

- Modified `wiced_bt_dev_read_tx_power()` to send the HCI_Read_Transmit_Power_Level HCI command
- Added a new API, `wiced_bt_set_transmit_power_range()` (See API documentation for details)
- Added a fix to disallow signed write command on EATT channel as per BT Core Spec
- Improved Doxygen API content and formatting

Bluetooth® Linux code examples and supported chips

3 Bluetooth® Linux code examples and supported chips

Table 3 Bluetooth® Linux code examples and supported chips

Code example	Feature demonstration	Supported chips
Linux Bluetooth® Find me	Demonstrates the Find me profile, which defines the behavior when a button is pressed on one device to cause an alerting signal on a peer device.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373
Linux Bluetooth® hello sensor	Demonstrates GATT database and device configuration initialization, sending data to the client and processing write requests from the client	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373.
Linux Bluetooth® Wi-Fi onboarding	Demonstrates a feature that enables devices to connect to a Wi-Fi access point without requiring a physical interface. This eliminates the need for cumbersome cables and allows for communication of Wi-Fi information through Bluetooth®, including SSID, password, control command, and device Wi-Fi status, providing a true wireless solution.	CYW55573/CYW55572/CYW55571, CYW43022.
Linux Bluetooth® Headset	Multiple profile CE, which demonstrates the use cases and abilities of audio-related functions like A2DP, AVRCP CT, HFP. It also supports GFPS, making it easier for the user to pair devices with as little user interaction as possible.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571, CYW43022, CYW43439, CYW4373.
Linux Bluetooth® Wake on LE	Demonstrates use of APCF function to achieve Wake on LE.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® LE Audio CIS Source	Implement the unicast source application using BTSTACK and LE Audio profile library.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® LE Audio CIS Sink	Implement the unicast sink application using BTSTACK and LE Audio profile library.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® LE Audio BIS Source	Demonstrates the ability of LE Audio broadcast.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571
Linux Bluetooth® LE Audio BIS Sink	Demonstrates the ability to receive LE Audio broadcast.	CYW55513/CYW55512/CYW55511, CYW55573/CYW55572/CYW55571

Bluetooth® Linux code examples and supported chips

3.1 Bluetooth® libraries

Table 4 Bluetooth® libraries

Libraries and middleware	Library details
bluetooth-linux	The bluetooth-linux is the adaptation layer (porting layer) between the Linux Bluetooth® application code example and Infineon's BTSTACK running on the Linux-based platforms. The porting layer provides Bluetooth® Stack initialization and implements platform interfaces to provide OS and memory services, enabling communication between the BTSTACK and the Bluetooth® controller.
bt-audio-profiles	Contains source and header files for A2DP, AVRCP, HFP, and SPP profiles.
btsdk-gfps	Contains source and header files for Google Fast Pairing service.
btstack	BTSTACK is Infineon's Bluetooth® Host Protocol Stack implementation. The stack is optimized to work with Infineon Bluetooth® controllers. The BTSTACK supports Bluetooth® BR/EDR and Bluetooth® LE core protocols.
le-audio-profiles-linux	Provides implementation of various LE Audio profiles, GATT interface utility for LE Audio code example, audio module library, and ISOC data handler interface.
fw	Folder contains various firmware files for the AIROC™ combo chip (CYW55513/2/1, CYW55573/2/1, CYW43022, CYW43439, CYW43455 and CYW4373).

Tools and documentation

4 Tools and documentation

Please refer to the following documents for more details:

Tools	Scope
Manufacturing Bluetooth® Test tool (MBT)	The manufacturing Bluetooth® Test tool (MBT) is used to test and verify the RF performance of the Infineon Bluetooth® Classic and Bluetooth® Low Energy devices on Linux platforms. See MBT on GitHub .
AIROC™ Bluetooth® Test and Debug tool	AIROC™ Bluetooth® Test and Debug tool is a GUI tool for testing and debugging Infineon Bluetooth® devices. AIROC™ Bluetooth® Test and Debug Tool connects to the Bluetooth® devices at the HCI protocol layer and currently supports HCI UART and HCI USB transport interfaces. The tool allows users to send Bluetooth® HCI commands and receive Bluetooth® HCI events from the Bluetooth® controller of the connected devices. Download the tool from the AIROC™ Bluetooth® Test and Debug webpage.

Glossary

Glossary

A2DP

advanced audio distribution profile

ACL

asynchronous connection oriented

ADI

Advertisement data information

AEC-CCM

advanced encryption standard-counter with cipher block chaining message authentication code

API

application programming interface

APCF

advertising packet content filter

AVRCP

audio/video remote control protocol

BIS

broadcast isochronous stream

BR

basic rate

BTSTACK

Bluetooth® Stack

CIS

connected isochronous stream

CSSA

Coding Scheme Selection for Advertising

EDR

enhanced data rate

GATT

generic attribute profile

HCI

host controller interface

Glossary

HFP

hands-free profile

ICS

implementation conformance statement

IOP

interoperable

ISOAL

isochronous adaptation layer

ISOC

isochronous channels

IUT

instrument under test

LL

link layer

LMP

link manager protocol

MIC

message integrity check

MOS

mean opinion score

OS

operating system

PAwR

periodic advertising with responses

PDU

protocol data unit

Rx

Reception

SCO

synchronous connection oriented

SMP

security manager protocol

Glossary

SPP

serial port profile

TCRL

test case reference lists

Tx

transmission

WBS

wide band speech

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Email:

erratum@infineon.com

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